Homework - Week 1 Codding Challenge - Github and Claude Code Skills Development

To lay the groundwork for production-grade development, I first set up my local environment and aligned it with the structure of the TrackPoli repository. I configured Claude Code Desktop (CCD) on Linux, connected it directly to GitHub, and linked it to the project while aligning my local file structure accordingly. In parallel, I built a solid understanding of core engineering concepts such as Gates, Repositories, CI Checks, Branches, and Application Surfaces, enabling me to navigate project setups and pull requests independently. Rather than focusing solely on execution, I aimed to understand the full stack required for the first homework/ AI Development Residency, in general. I clarified how backend systems, frontend frameworks, security gates, and CI checks interact, and specifically distinguished between the rigid core infrastructure (the “hard structure”) and the flexible, dynamic application layers (the “soft structure”).

Instead of using AI merely to generate quick code, I applied structured prompting to deepen my domain knowledge and guide my execution more effectively. I provided the model with comprehensive context about the AI Development Residency, my explicit goals, and relevant project references such as github/trackpoli, creating a grounded foundation for the work. I then used the agent to obtain conceptual feedback and alternative formulations, which helped me synthesize my notes into a clear and unified narrative. Through this process, I moved from basic prompt interaction toward a structured, agentic architecture: establishing repeatable workflows and explicit constraints that lead to more predictable and reliable outcomes.

To figure out what actually makes a good skill document, I compared outputs from different AI agents using the prompt: *"What further skills beside UX, Accessibility, Security would create a cohesive skills set for an enterprise web app and provide reasoning."* Through this comparison, I learned that a good skill document isn't just a list of general topics. Instead, it needs to be concise, focus on solving specific real-world problems, and set clear rules that the AI can follow consistently every time.

From comparing the AI responses, I also learned that while UX, Accessibility, and Security are important, an enterprise app needs skills focused on Performance, Observability, and Data Management to run smoothly. Without these, apps can quickly become slow, buggy, or expensive to host.

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